Evaluating Multi-Angle Dataset on Embedded Vision System for Reverse Vending Machine

ENGR 895 Thesis Project

by

Dung Huynh

Guided by Dr. Xiaorong Zhang

Supported by Ayushi Patel, Dr. Zhuwei Qin

August 2025

Table of Content

I. INTRODUCTION ……1
II. SYSTEM ARCHITECTURE AND DESIGN……1
A. System Overview
B. Hardware Components
III. DATA COLLECTION AND ANALYSIS……4
A. Survey methodology and findings
B. Dataset
C. Data Preprocessing
IV.MODEL……..8
A. Model Selection
B. Model Training
V. SYSTEM IMPLEMENTATION AND INTEGRATION……..9
A. Deployment on Embedded System
B. Validation
VI. EVALUATION……..10
A. Position
B. Light
C. Background
VII. NON-TECHNICAL ANALYSIS……12
A. Survey Insights
B. Stakeholder Recommendations
VIII. CONCLUSIONS……...15
A. Project Summary
B. Future Research Directions
IX. REFERENCES……16
X. APPENDICES……..17
A. Survey Instruments
B. Additional Materials

I. INTRODUCTION

According to the Environmental Protection Agency (EPA), in 2018 only 25% of glass, 8.7% of plastics, and 34% of metals generated in the United States were recycled [1], while the remainder was transported to landfills. By 2025, plastic recycling rates were still below 10%. These numbers demonstrated environmental concerns as well as significant losses in economic value. In Maximizer Market Research, the U.S. recycling and waste services market was about to reach USD 98 billion by 2030 [2]. The gap between low recycling performance and industry growth highlighted the need for technological solutions that could improve individual recycling behaviors and capture untapped market potential.

This project introduced the development of a Reverse Vending Machine (RVM) system for classifying recyclables as plastic, glass, and aluminum, using computer vision and deep learning models deployed on a Raspberry Pi. It also evaluated how environmental conditions and features influenced classification performance.

A survey of 77 participants was conducted to understand recycling behaviors, motivations, and preferences for smart recycling centers as the objective for this project. The collected responses were considered to ensure the system design was efficient, secure, and user-friendly. This paper presented the completed pipeline, including data collection, preprocessing, model training, evaluation, and real-time inference integration.

II. SYSTEM ARCHITECTURE AND DESIGN

A. System Overview

The proposed architecture for this project was divided into three main parts: the object and camera, the Raspberry Pi, and the computer (Inspiron 15 551). (figure 1)

First, the camera captured the object—such as a bottle or can—using the Raspberry Pi Camera Module 3 and transmitted the image to the Raspberry Pi through a ribbon cable and a CSI

(Camera Serial Interface) connector. A training data collection program on the Raspberry Pi designed to store these captured images.

The collected images were then uploaded to Google Drive for shared access between the Raspberry Pi and the computer. Preprocessing was performed on the Inspiron 15 551 using a resizer application. Model training and testing were conducted on Google Colab using transfer learning for efficiency. Once trained and tested, the models were exported into .jit/.pt format and uploaded back to Google Drive.

On the Raspberry Pi, the trained models were loaded, executed for prediction, and used to display the final classification decision.

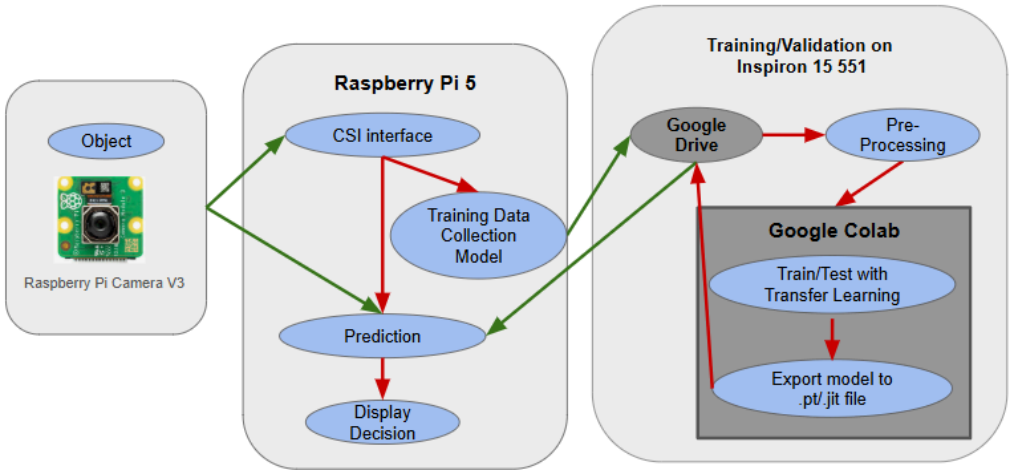


Figure 1: Proposed System Architecture

B. Hardware Components

1. Raspberry Pi 5

The Raspberry Pi 5 was selected for this project because of its large library support, strong computational power, and affordability. It had a quad-core CPU and GPU capable of training deep learning models such as MobileNetV2 in real time, making it suitable for embedded vision applications. In addition, the Raspberry Pi 5 had supported camera integration and GPIO

connections. Raspberry pi also had a large, active developer community that provided helpful resources for deployment and troubleshooting.

2. Camera Module

The Camera Module 3 was chosen because of its compatibility with the Raspberry Pi and its consistency in capturing images. It also supported the deployment and live testing for system classification in the later stage.

III. DATA COLLECTION AND ANALYSIS

A. Survey methodology and findings

A survey was conducted to understand user behavior, motivations, feature preferences, and the limitations or trade-offs participants were willing to accept when it comes to recycling. It also explored where users prefer to recycle and what would encourage them to recycle more. The survey included 11 multiple-choice or rating questions, along with 1 optional open-ended question for additional feedback.

- Questions 1 and 2 explored users' current recycling habits and patterns.
- Questions 3, 4, and 5 focused on recycling motivations.
- Questions 6 and 8 identified important features to guide the design and development priorities.
- Questions 7 and 9 addressed placement and integration of RVM
- Questions 10 and 11 assessed system limitations and trade-offs

The survey result indicated a low percentage of responders actually recycled regularly, particularly outside of their home. However, a high number of them show willingness to recycle if there were a reverse vending machine that offered automatic sorting and provided conveniences in terms of time and cleanliness.

B. Dataset

The dataset was created using two primary sources: (1) internet-based image collection and (2) self-collected images using the Raspberry Pi Camera Module 3.

1. Internet Collection

Multiple datasets were available online and collected such as *TrashNet*, *Kaggle Garbage Classifier*, *Kaggle Plastic Bottle*, and *Recycle32*. Images were selected based on three material categories: plastic, glass, and aluminum. Additional background images were collected separately online. However, the collected images distribution was imbalanced, additional images were acquired manually using a Samsung Galaxy Note 10. Photos were taken with clean backgrounds, wood, white, or cement, to maintain focus on the objects and from multiple angles to increase variation.

Since reverse vending machines (RVMs) typically operated in controlled environments with limited object positions, the variability in resolution and angles from online datasets was not practical. This limitation led to the creation of a custom dataset more applicable for RVM scenarios.

2. Self-Collected Dataset

2.1 Data collecting Methodology

At first, objects were captured standing upright on a cardboard background for side view. Lighting was collected in an ambient light room environment with two variations: indirect light from a nearby source to the object, and another aimed directly at the object. This setup allowed observation of how reflections influenced recognition. The side view provided better area exposure of each object, which supports material classification.

Since the objects in real RVM were deposited horizontally, this side view still did not replicate how items would be inserted truthfully. Therefore, a top-and-bottom dataset was created. In this setup, objects were placed on an elevated surface to simulate a conveyor belt, and images were captured from top and bottom of the object, under both lighting variations.

2.1a Visual Shape Analysis

Glass and plastic sometimes shared similar visual features, making classification difficult. Aluminum object's, on the other hand, were more distinct. Analyzing top and bottom views, revealed differentiating features: aluminum cans displayed unique top and bottom circular features; glass bottles generally had a ridged ring around the outer base, designed for stability and safety, while plastic bottles exhibited more varied common base textures. These features provided potential keys for material differentiation.

2.1b Lighting and Background Variations

Another dataset was created using a black plastic base and cardboard side walls to examine whether a darker base would improve reflection classification. Images were captured from top and bottom views using direct lighting.

2.1c Object Diversity and Image Count

To ensure diversity, the dataset includes 10 distinct objects for each material type, each was captured 30 times with variation in distance (close to far), cap presence (with or without cap), and different sides exposed.

2.2 Data Acquisition for Training and Testing Dataset

The data was captured by Raspberry Pi 5 and Raspberry Pi Camera Module 3 with a custom Python script. It used the following keyboard inputs during the process:

- space – capture and save the current image
- n – switch to the next material class
- q – quit the capture session

Before entering capture mode, the script prompted for the background type, object position, and lighting condition. This metadata was used to organize and label images systematically for the following format: [class]_[background]_[position]_[lighting].jpg with the following description:

- Position (p): p1 = side view; p2 = top and bottom views
- Background (b): b1 = cardboard; b2 = black bottom with cardboard side
- Lighting (l): l1 = InDirect light; l2 = Direct light

Images were organized based on background type to simplify dataset management. Since the Raspberry Pi Camera Module 3 captures images in RGB format, but OpenCV displayed in BGR, a color conversion was applied for consistency.

The dataset was splitted in 80% training and 20% testing. Table 1 showed the summary of the collected dataset which separate scenarios for each condition of position, lighting and background. Table 2 summarized the details of each dataset

Comparison	Dataset abv	View	Lighting	Background
Position (p)	**P1**	Side (p1)	InDirect + Direct (l1 + l2)	Cardboard (b1)
	P2	Top & Bottom (p2)		
Lighting (l)	**L1**	Side + Top& Bottom (p1+ p2)	InDirect (l1)	Cardboard (b1)
	L2		Direct (l2)	
Background (b)	**B1**	Top & Bottom (p2)	Direct (l2)	Cardboard (b1)
	B2			Black +Cardboard (b2)

Table 1 : Collected Dataset name and conditions for comparison

Background	Views	Lighting Condition	Objects / Category	Images / Class	Total Images / class	Total Images	
Cardboard (self) (b1)	Side view (p1)	InDirect (I1)	5	150	450	~1800	~3600
		Direct (I2)	10	300			
	Top&bottom view (p2)	InDirect (I1)	5	150	450	~1800	
		Direct (I2)	10	300			
Black - Cardboard (self)(b2)	Top&bottom view (p2)	Direct (I2)	10	300	300	~1200	~1200

Table 2 : Dataset Detail

2.3 Validation Dataset

A validation dataset of approximately 5% of the total dataset was created. It consisted of 2 objects for each material class at three distances to the camera. The figure below showed the validation dataset of top&bottom view, direct light and cardboard.

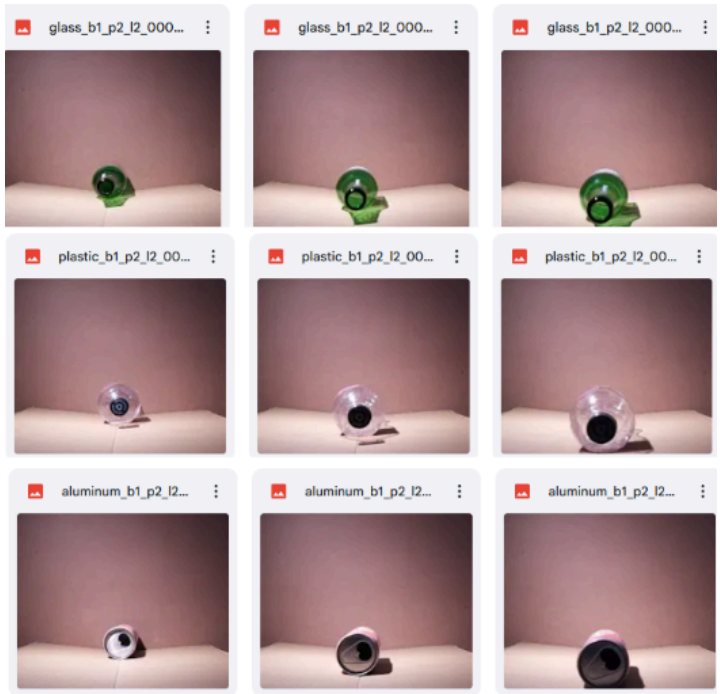


Figure 2: Validation dataset for top&bottom view

C. Data Preprocessing

All images were preprocessed using the Photo Resizer application to 224×224 pixels to match the MobileNetV2 model input requirement. Manual preprocessing ensured consistency in labeling, and allowed easier editing of mislabeled files

IV. MODEL

A. Model Selection

In this project, MobileNetV2 was selected due to its efficiency and reliability for image classification tasks. MobileNetV2 was designed for embedded devices and demonstrated a high accuracy in object classification, as well as recyclable objects [3] compared to the other model as ResNet50 or MobileNetV1. It also achieved nearly 97% accuracy in plastic classification in prior studies [4].

Pretrained on the ImageNet dataset, MobileNetV2 offers a strong balance between accuracy, inference speed, and model size—making it a good fit for small datasets and deployment on devices such as the Raspberry Pi 5. Compared to MobileNetV3, MobileNetV2 offers faster inference, simpler architecture, and better suitable option for deployment on Raspberry Pi . While MobileNetV3 can achieve slightly higher accuracy, its model complexity and resource demands make MobileNetV2 a more practical and efficient choice for this project [5]

B. Model Training

Transfer learning was applied by leveraging the pretrained weights and parameters of MobileNetV2 to accelerate training and improve performance, eliminating the need to develop a custom architecture from scratch. During preprocessing, input images were normalized using the standard ImageNet mean and standard deviation values: *mean = [0.485, 0.456, 0.406]* and *std = [0.229, 0.224, 0.225]*. These values ensured compatibility with the original pretrained model. The dataset was divided into 80% for training and 20% for testing, with fixed splits used

to maintain consistency during evaluation. Model development was informed by Daniel Bourke's *Zero to Mastery: Learn PyTorch for Deep Learning*, which provided guidance for code structure and model implementation. [6]

The model was trained using a cross-entropy loss function with a learning rate of 0.001 and achieved over 96% accuracy across all train and test sets. Misclassifications occurred in a few cases, primarily due to variability in the input data rather than deficiencies in the model architecture.

For deployment, the trained model was converted to a TorchScript .jit format to execute on Raspberry Pi. This format contributed to faster inference and lower deployment risk. The model was trained on Google Colab using a Tesla T4 GPU, with a batch size of 32 over 20 epochs, completing in under five minutes.

V. SYSTEM IMPLEMENTATION AND INTEGRATION

A. Deployment on Embedded System

The trained model was deployed on the Raspberry Pi connected to Raspberry Pi Camera Module 3 for live testing. The testing was performed in the ambient light environment with limited control of conditions. The setup replicated the environments used during data collection, with adjustments in lighting and background.

During live prediction, a Region of Interest (ROI) was applied to isolate the object and minimize the background noise. It helped to improve the prediction accuracy. The system displayed the classification result on top of ROI along with the inference time. If the confidence was more than 80%, the predicted class and inference time were shown. Otherwise, it displayed “Analyzing” with the predicted class and its confidence.

B. Validation

Validation testing was performed by reusing the same objects, backgrounds, approximate timings, placements, and lighting conditions as conducted during validation dataset creation.

The model showed close alignment with the offline predictions confirming that the trained model maintained consistency when deployed.

VI. EVALUATION

The accuracy of the train, test and validation dataset offline were summarized in Table 3. The inference time was generated approximately from 0.1s to 0.2

Dataset Name	Accuracy (Train) (%)	Accuracy (Test) (%)	Inference Time (s)	Accuracy (Validation) (%)
P1	98.09	97.12	0.238	88.46
P2	**99.15**	**98.82**	0.236	90.98
L1	97.72	**99.65**	0.186	50
L2	**99.15**	98.86	0.171	88.10
B1	97.92	**99.31**	0.189	85.71
B2	**98.52**	98.26	0.192	75

Table 3 Comparison of Accuracy for Offline and Live Result

A. Position

The offline results showed that the top&bottom position achieved slightly better accuracy (99.15%) compared to the side position(98.09%) in training. Both of the positions exceeded 97% accuracy in testing. In the validation testing, the top&bottom position also outperformed side view (90.98% and 88.46%)

When demonstrating the prediction between classes, the following result was recognized for each material in table 4. Aluminum was predicted consistently under both positions. Glass showed mixed results with more correct than incorrect classification. Plastic was more accurately classified in top&bottom position than side position.

Position/Material	Aluminum	Glass	Plastic
Side	✅	✅✅❌	✅✅❌
Top&Bottom	✅	✅✅❌	✅

Table 4: Material comparison for Position condition with Direct Lighting, Cardboard background

In top&bottom feature analysis, the bottom of the object was classified better than the top. The tops of glass and plastic showed similar shapes and visual feature, while the bottoms provided distinguishing features that are more reliable to differentiate the materials

B. Light

The offline results showed the indirect light offered lower training accuracy than direct lighting, though the test accuracy was slightly higher. In the validation testing, a significant accuracy dropped in the indirect lighting, likely due to the uncontrolled ambient lighting environment and limited dataset variety

In the live classification, the following was explored in table 5 . Aluminum maintained overall high accuracy regardless of lighting condition. Glass performed better under direct light with some misclassification. Plastic accuracy was much lower in the InDirect lighting compared to the Direct lighting.

Light/Material	Aluminum	Glass	Plastic
InDirect	✅	❌	❌
Direct	✅	✅✅❌	✅

Table 5: Material comparison for Lighting condition with top&bottom position, cardboard background

Overall, the Direct lighting showed the best result for all three materials.

C. Background

Training and testing accuracies were above 98% for both cardboard and black-cardboard backgrounds (training for cardboard was 97.92%), though validation accuracy was lower: 85.71% for cardboard and 75% for black-cardboard.

The material analysis indicated that aluminum and plastic were classified more accurately with cardboard backgrounds, while glass performed better on black-cardboard backgrounds, likely due to reflections improving detection of transparent surfaces.

Background/Material	Aluminum	Glass	Plastic
Cardboard	✅	❌	✅
Black-Cardboard	❌	✅	❌

Table 6 : Material comparison for Background with direct light, top&bottom position

VII. NON-TECHNICAL ANALYSIS

A. Survey Insights

Key Findings from 77 Respondents:

- 1. Recycling Frequency: showed that a significant portion of people still do not recycle regularly.
  26 participants said they rarely or never recycle, or only recycle once or twice a year.
  10 recycle every 2 to 3 months.
  21 recycle daily, while the remaining 20 recycle weekly or monthly.

- 2. Amount Recycled per Visit
  28 participants reported recycling between 50 to over 100 items per visit.
  11 recycle fewer than 10 items.
  20 said they don't keep count.
  17 recycle between 11 to 50 items.

- 3. Why People Recycle *(up to 2 selections allowed)*

  54 people cited environmental concerns
  31 pointed to habit or upbringing
  25 chose financial rewards
  21 said the presence of nearby recycling bins motivated them
  12 said recycling makes them feel good
  10 mentioned peer influence or community rules

- 4. Preferred Types of Rewards

62 respondents preferred cash
28 preferred coupons
13 would donate rewards to charity
11 would like redeemable points
10 preferred entering a raffle
9 said they didn't care about rewards

- 5. Financial Incentive Expectations

  23 participants said they don't need financial rewards to recycle
  18 said they would need more than $20 per month
  11 would recycle for any amount
  25 said $1 to under $20 would be acceptable

- 6. Feature Preferences
  People expressed interest in app features in addition to simplicity and ease of use.
  57 would recycle more if a machine could automatically sort items
  18 would recycle more if the process was fast and easy

- 7. Best Locations for Recycling Machines *(up to 2 selections allowed)*

  71 people selected grocery stores or supermarkets
  37 selected parks and recreational areas
  Apartments, schools, offices, and public buildings each received more than 30 votes

- 8. App Use for Tracking Recycling

  33 would use an app to track their recycling
  27 would use it if it was easy to use
  17 preferred not to use an app

- 9. Important Machine Qualities
  Cleanliness and time-saving convenience were identified as must-have features.
  Cashback and other rewards were also important considerations.

- 10. Machine Speed Acceptability

  40 participants would still use the machine regularly, even if it could only sort 5–10 items per minute
  13 would use it less often
  7 would use it only if they had a small number of recyclables
  7 would use it if there was no wait line
  Only 4 would avoid using it altogether

- 11. Personalization and Accessibility
  Most users said they would use the machine if it could accommodate their preferences, such as:

wait time display
immediate reward
drop-off with reward received later
multiple machines available
convenient locations
Only 6 respondents said they would not use the system.

- 12. Additional Feedback *(optional responses)*
  Participants feedback the importance of making the machine accessible to all, including support for multiple languages, ease of use for elderly and disabled individuals, clear graphic or instructions for recyclable items, and overall system security. Many also mentioned that timing and machine availability were recurring concerns.

B. Stakeholder Recommendations

A short interview with Olyns Cube users and an operator provided further insights:

1. Customers

- Customer 1: began recycling after the Olyns machine was installed in the city, during daily walks, in the park, and from coworkers who did not recycle. The customer visited the Olyns Cube twice a week and earned over $600 in the past two years. The motivation was from environmental concern, and personal benefit
- Customer 2: recycled weekly for one year, mainly plastic bottles. The motivator is a convenient location, near a shopping center.
- Customer 3 had a negative first experience due to the long wait, and full bin. The customer stored recyclables in a garage for nearly a year. Decided to try again, the RVM was still unavailable due to a software update. If the experience would not improve, the customer indicated a possibility of discarding recyclables in the landfill.

Based on these interviews, RVMs did encourage people to recycle. It demonstrated strong potential to build and support consistent recycling habits.

2. Operator Interview and Observations

The operator managed multiple units independently with busiest units requiring up to 6 pickups per day. The operator stated that glass bins were heavy and unsafe to handle, while glass bottles tended to break and bags were prone to tearing.

In observation, residual liquids from recyclables created odor and sanitation concerns.These insights from the users and operator were valuable for future improvements in machine design and operational safety.

VIII. CONCLUSIONS

A. Project Summary

In conclusion, this project demonstrated the development of an embedded vision system for recyclable bottle classification, with a completed pipeline including: data collection, preprocessing, model training, evaluation, and deployment on Raspberry Pi for real-time inference. The model provided optimal accuracy in top&bottom position, and direct lighting with Cardboard background for aluminum and plastic, and Black-Cardboard background for glass. This background application would be beneficial if needed to classify single material for RVM. The project also incorporated survey and stakeholder feedback, highlighting user motivations, design requirements, and operational challenges in real-world RVM systems

B. Future Research Directions

Limitations included dataset size and diversity with restricted environments, which contributed to lower live prediction accuracy compared to offline results. Future worked should focus on expanding dataset with greater diversity and size, integration of object localization and detection technique as sensor and ROI refinement, adding hardware features such as weight sensors and Universal Product Code (UPC) code scanning for material identification, as well as investigating multi-object classification within the same frame to enhance RVM practicality.

IX. REFERENCES

[1] United States Environmental Protection Agency, "Advancing sustainable materials management," EPA, [Online]. Available: https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/advancing-sustainable-materials-management. [Accessed: 03-Jul-2025].

[2] Maximize Market Research, "U.S. Waste Management and Recycling Market: Industry Analysis and Forecast (2024–2030) by Waste Type, Disposal Method, End User and Region," Maximize Market Research, [Online]. Available: https://www.maximizemarketresearch.com/market-report/u-s-waste-management-and-recycling-market/222524/. [Accessed: 03-Jul-2025].

[3] J. Caballero, F. Vergara, R. Miranda, and J. Serracín, "Inference of recyclable objects with convolutional neural networks," unpublished. [Online]. Available: https://revistas.utp.ac.pa/index.php/ric. [Accessed: 03-Jul-2025].

[4] S. Kunwar, B. R. Owabumoye, and A. S. Alade, “Plastic waste classification using deep learning: insights from the WaDaBa dataset,” arXiv Preprint, Dec. 2024, arXiv:2412.20232

[5] A. Howard, M. Sandler, G. Chu, L. C. Chen, B. Chen, M. Tan, and Q. V. Le, "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1314–1324. doi: 10.1109/ICCV.2019.00140.

[6] D. Bourke, "Learn PyTorch for deep learning," [Online]. Available: https://www.learnpytorch.io/. [Accessed: 03-Jul-2025].

[7] TOMRA, "TOMRA reverse vending technologies," TOMRA, [Online]. Available: https://www.tomra.com/reverse-vending/our-offering/tomra-reverse-vending-technologies. [Accessed: 02-June-2025].

[8] Statista, "Plastic packaging municipal waste recycled in the U.S. 1980–2018," Statista, [Online]. Available: https://www-statista-com.jpllnet.sfsu.edu/statistics/1228181/us-plastics-container-packaging-waste-recycled/. [Accessed: 05-May-2025].

[9] B. D. Carolis, F. Ladogana, and N. Macchiarulo, "YOLO TrashNet: Garbage detection in video streams," in *Proc. 2020 IEEE Conf. Evolving and Adaptive Intelligent Systems (EAIS)*, Bari, Italy, 2020, pp. 1–7. doi: 10.1109/EAIS48028.2020.9122693.

[10] R. Aarthi and G. Rishma, "A vision based approach to localize waste objects and geometric features extraction for robotic manipulation," *Procedia Comput. Sci.*, vol. 218, pp. 1342–1352, 2023. doi: 10.1016/j.procs.2023.01.113.

[11] Md. N. Nahiduzzaman, Md. F. Ahamed, M. Naznine, Md. J. Karim, H. B. Kibria, M. A. Ayari, A. Khandakar, A. Ashraf, M. Ahsan, and J. Haider, "An automated waste classification system using deep learning techniques: Toward efficient waste recycling and environmental

sustainability," *Knowl.-Based Syst.*, vol. 310, 113028, 2025. doi: 10.1016/j.knosys.2025.113028.

[12] Alan Sun, "Smart Trash: Object detection and classification for waste sorting," GitHub, 2022. [Online]. Available: https://github.com/alansun17904/smart-trash. [Accessed: 20-Mar-2025].

[13] Y. Ruparel, L. Chaudhary, and A. Rathod, "AI-based smart bin for efficient and sustainable waste classification," *World J. Environ. Eng.*, vol. 9, no. 1, pp. 1–6, 2024. [Online]. Available: https://pubs.sciepub.com/wjee/9/1/1/index.html.

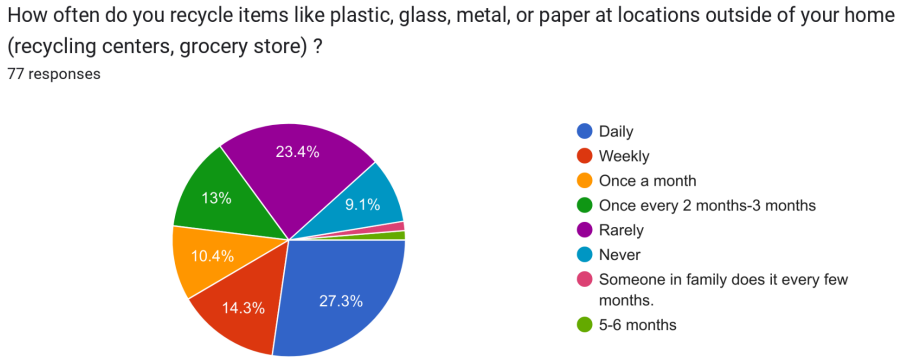
[14] Y. Ruparel, L. Chaudhary, and A. Rathod, "AI-based smart bin for efficient and sustainable waste classification," *World J. Environ. Eng.*, vol. 9, no. 1, pp. 1–6, 2024. [Online]. Available: https://pubs.sciepub.com/wjee/9/1/1/index.html.

[15] A. King, "Recyclable and household waste classification," Kaggle, 2022. [Online]. Available: https://www.kaggle.com/datasets/alistairking/recyclable-and-household-waste-classification.

X. APPENDICES

A. Survey Instruments

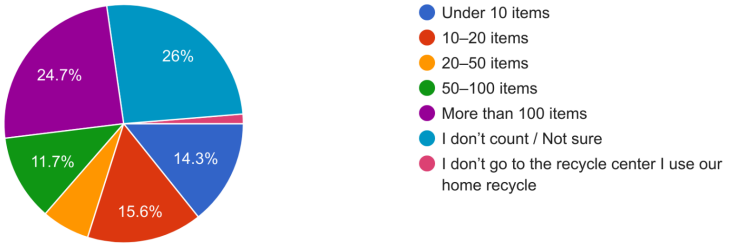
Question 1



Question 2

On average, how many bottles or cans do you bring to recycle at one time? This helps us understand the range of users — whether you're dropping off a few items or bringing a full load!

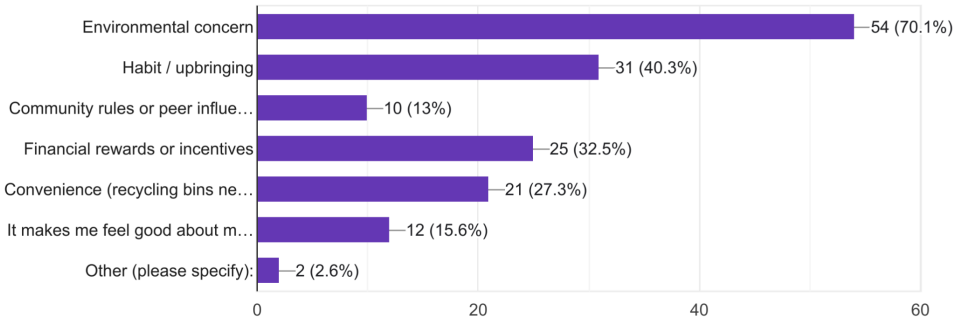
77 responses



Question 3

What motivates you to recycle the most? (Select up to 2)

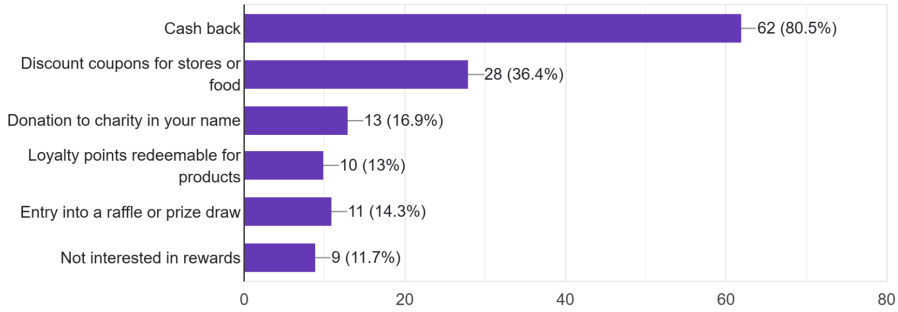
77 responses



Question 4

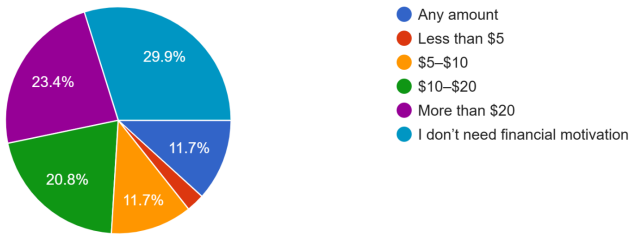
If you could earn rewards for recycling, which of the following would interest you the most? (select up to 2)

77 responses



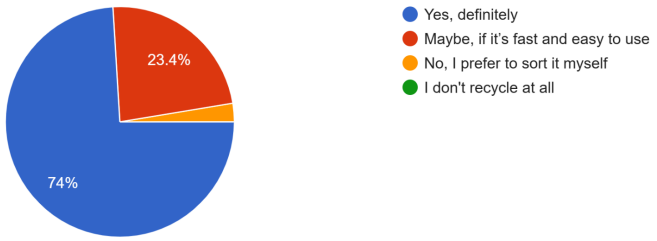
Question 5

How much would you need to earn per month from recycling to be motivated to do it regularly?
77 responses



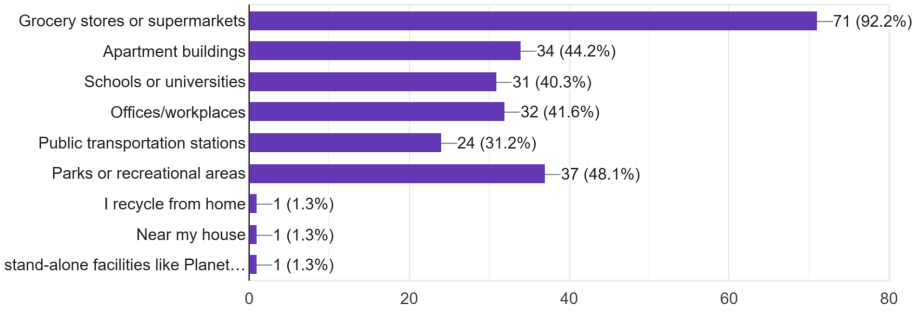
Question 6

Would you be more likely to recycle if a smart machine could sort your waste automatically?
77 responses



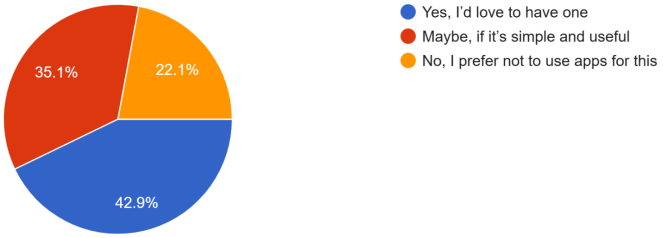
Question 7

Where would you prefer to find Smart Recycling Centers? (Select all that apply)
77 responses



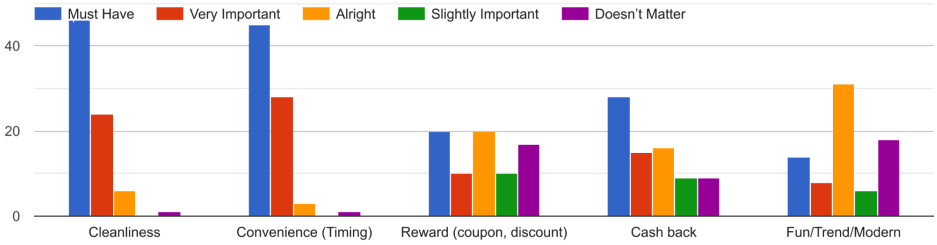
Question 8

Would you use a mobile app to track your recycling habits and rewards?
77 responses



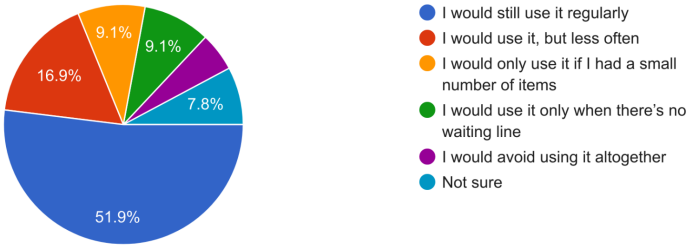
Question 9

Rate how important each of the following features is for you when using a Smart Recycling Station.



Question 10

If a Smart Recycling Station could only accept 5–10 items per batch, and each batch takes about 1 minute to process (Imagine standing there for abo...ner your home becomes as you free it from clutter!
77 responses

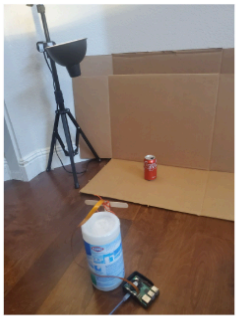


Question 11

What would encourage you to use a recycling machine that processes only 5–10 items per minute?
77 responses

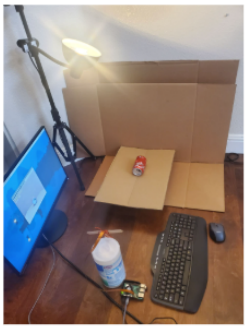
B. Additional Materials

Setup Images

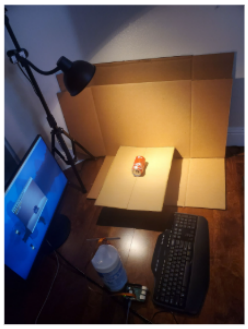


Side (p1)

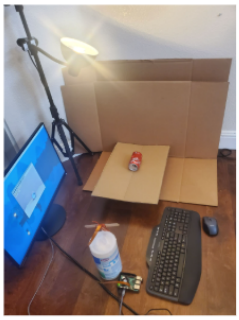
Top&bottom (p2)



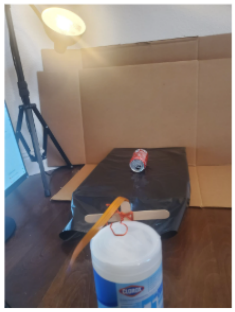
InDirect Light (l1)



Direct Light (l2)



Cardboard (b1)



Black-Cardboard (b2)